

# Independence

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**Definition 0.1** (Statistical independence). Random variables  $X_1, \dots, X_n$  are **statistically independent** if, for all sets of real numbers  $A_1, \dots, A_n$ , the **probability** that every  $X_i$  falls in its set is the product of the individual probabilities:

$$\Pr(X_1 \in A_1, \dots, X_n \in A_n) = \prod_{i=1}^n \Pr(X_i \in A_i)$$

*Remark.* We write  $X \perp\!\!\!\perp Y$  for “ $X$  and  $Y$  are independent”. The symbol  $\perp\!\!\!\perp$  is essentially  $\prod$  upside-down, which can remind you of the definition.

**Theorem 0.1** (Independence of discrete random variables: the joint PMF factors). *Discrete random variables  $X_1, \dots, X_n$  are **statistically independent** if and only if their **joint PMF** factors into the product of their PMFs: for all real numbers  $x_1, \dots, x_n$ ,*

$$\Pr(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n \Pr(X_i = x_i)$$

## Proof

*Proof.* **Only if.** Take  $A_i = \{x_i\}$  for each  $i$  in Definition 0.1.

**If.** Let  $A_1, \dots, A_n$  be sets of real numbers, and let  $C$  be the set of tuples  $(x_1, \dots, x_n)$  with each  $x_i \in A_i \cap \mathcal{R}(X_i)$ . Each **range**  $\mathcal{R}(X_i)$  is countable, so  $C$  is countable. Each  $X_i$  takes its values in  $\mathcal{R}(X_i)$ , so the event  $\{X_1 \in A_1, \dots, X_n \in A_n\}$  is the disjoint union of the events  $\{X_1 = x_1, \dots, X_n = x_n\}$  over  $(x_1, \dots, x_n) \in C$ :

$$\begin{aligned}
\Pr(X_1 \in A_1, \dots, X_n \in A_n) &= \sum_{(x_1, \dots, x_n) \in C} \Pr(X_1 = x_1, \dots, X_n = x_n) && \text{(countable additivity over the disjoint event set } C \text{ of non-negative real numbers)} \\
&= \sum_{(x_1, \dots, x_n) \in C} \prod_{i=1}^n \Pr(X_i = x_i) && \text{(the joint PMF factors)} \\
&= \prod_{i=1}^n \sum_{x_i \in A_i \cap \mathcal{R}(X_i)} \Pr(X_i = x_i) && \text{(a sum over the product set } C \text{ of non-negative real numbers)} \\
&= \prod_{i=1}^n \Pr(X_i \in A_i) && \text{(countable additivity over the disjoint event set } C \text{ of non-negative real numbers)}
\end{aligned}$$

□

**Example 0.1** (Two fair coin flips). Flip two fair coins, and let  $X_1$  and  $X_2$  indicate heads on the first and second flip. Each of the four outcomes has probability  $1/4$ , so for example:

$$\begin{aligned}
\Pr(X_1 = 1, X_2 = 1) &= \frac{1}{4} \\
&= \frac{1}{2} \cdot \frac{1}{2} \\
&= \Pr(X_1 = 1) \Pr(X_2 = 1)
\end{aligned}$$

and the same factorization holds for the other three pairs of values, so  $X_1 \perp\!\!\!\perp X_2$  by Theorem 0.1. In contrast,  $X_1$  and the total number of heads,  $X_1 + X_2$ , are not independent:  $\Pr(X_1 = 0, X_1 + X_2 = 2) = 0$ , but

$$\begin{aligned}
\Pr(X_1 = 0) \Pr(X_1 + X_2 = 2) &= \frac{1}{2} \cdot \frac{1}{4} \\
&= \frac{1}{8}.
\end{aligned}$$

**Theorem 0.2** (Independence with densities: the joint density factors). ***Continuous case.** Continuous random variables  $X$  and  $Y$  with [densities](#)  $p(X = x)$  and  $p(Y = y)$  are [statistically independent](#) if and only if the product  $p(X = x) p(Y = y)$  is a [joint density](#) of  $X$  and  $Y$ .*

***Mixed case.** A discrete random variable  $X$  and a continuous random variable  $Y$  with density  $p(Y = y)$  are statistically independent if and only if the product  $\Pr(X = x) p(Y = y)$  is a [joint density-mass function](#) of  $X$  and  $Y$ .*

*Remark.* The proof is beyond these notes' scope: it needs measure theory that the notes do not develop, to extend a density's integrals from intervals to general sets and to exchange the order of integration (Casella and Berger 2002; Billingsley 1995). The continuous case extends to  $n$  random variables with a joint density of all  $n$ . The statement says “a joint density” rather than “the joint density” because a density is not unique: changing it on a

set of zero area (zero length, in the mixed case) changes none of its integrals.

**Example 0.2** (The PMF form fails for continuous random variables). The factorization in Theorem 0.1 does not work as a definition of independence for continuous random variables: there, both sides are 0 at every point, so it would call every pair of continuous random variables independent. For instance, let  $X \sim \text{Uniform}(0, 1)$  ([uniform distribution](#)), whose density is 1 on  $[0, 1]$ , and let  $Y = X$ . For all real numbers  $x$  and  $y$ , the event  $\{X = x, Y = y\}$  is  $\{X = x\}$  if  $y = x$  and empty otherwise, so it has probability 0, because  $\Pr(X = x) = 0$  for the [continuous](#)  $X$ . Likewise

$$\begin{aligned}\Pr(X = x) \Pr(Y = y) &= 0 \cdot 0 \\ &= 0,\end{aligned}$$

so the PMF form holds. But  $X$  and  $Y$  are not independent:

$$\begin{aligned}\Pr(X \in [0, \tfrac{1}{2}], Y \in [0, \tfrac{1}{2}]) &= \Pr(X \in [0, \tfrac{1}{2}]) \quad (Y = X) \\ &= \int_0^{1/2} 1 \, dx \quad (\text{the density of } X \text{ is 1 on } [0, 1]) \\ &= \tfrac{1}{2} \quad (\text{integrate})\end{aligned}$$

while

$$\begin{aligned}\Pr(X \in [0, \tfrac{1}{2}]) \Pr(Y \in [0, \tfrac{1}{2}]) &= \tfrac{1}{2} \cdot \tfrac{1}{2} \\ &= \tfrac{1}{4}.\end{aligned}$$

**Definition 0.2** (Conditional independence). Random variables  $Y_1, \dots, Y_n$  are **conditionally independent** given a discrete random variable (or vector)  $\tilde{X}$  if, for every value  $\tilde{x}$  with  $\Pr(\tilde{X} = \tilde{x}) > 0$  and all sets of real numbers  $A_1, \dots, A_n$ , the conditional probability that every  $Y_i$  falls in its set is the product of the individual conditional probabilities:

$$\Pr(Y_1 \in A_1, \dots, Y_n \in A_n \mid \tilde{X} = \tilde{x}) = \prod_{i=1}^n \Pr(Y_i \in A_i \mid \tilde{X} = \tilde{x})$$

When  $\tilde{X}$  is continuous, every event  $\{\tilde{X} = \tilde{x}\}$  has probability 0, so the condition is stated with densities instead: for every  $\tilde{x}$  with  $p(\tilde{X} = \tilde{x}) > 0$  and all  $y_1, \dots, y_n$ ,

$$\frac{p(\tilde{X} = \tilde{x}, Y_1 = y_1, \dots, Y_n = y_n)}{p(\tilde{X} = \tilde{x})} = \prod_{i=1}^n \frac{p(\tilde{X} = \tilde{x}, Y_i = y_i)}{p(\tilde{X} = \tilde{x})}$$

where each  $p(\cdot)$  is a [joint density](#) (with probability masses in place of densities for any discrete  $Y_i$ , as in a [joint density-mass function](#)).

*Remark.* For continuous  $\tilde{X}$ , each ratio in the density form is the density of the  $Y$ 's under their conditional distribution given  $\tilde{X} = \tilde{x}$ , so both forms say the same thing: given  $\tilde{X} = \tilde{x}$ , the  $Y_i$ 's joint distribution factors.

**Example 0.3** (Two tests of the same patient). Let  $X$  indicate whether a patient has a disease, and let  $Y_1$  and  $Y_2$  indicate positive results on two tests whose errors are unrelated, so that  $Y_1$  and  $Y_2$  are conditionally independent given  $X$ . Suppose  $\Pr(X = 1) = 0.1$ , each test is positive with probability 0.9 if  $X = 1$  and with probability 0.1 if  $X = 0$ . Then, by the [law of total probability](#):

$$\begin{aligned}\Pr(Y_1 = 1, Y_2 = 1) &= (0.9)(0.9)(0.1) + (0.1)(0.1)(0.9) && \text{(condition on } X; \text{ factor given } X) \\ &= 0.081 + 0.009 && \text{(multiply)} \\ &= 0.09 && \text{(add)}\end{aligned}$$

but

$$\begin{aligned}\Pr(Y_1 = 1) &= (0.9)(0.1) + (0.1)(0.9) \\ &= 0.18,\end{aligned}$$

so  $\Pr(Y_1 = 1) \Pr(Y_2 = 1) = 0.0324 \neq 0.09$ : the tests are conditionally independent given  $X$ , but not independent.

**Example 0.4** (Two coins, given their total). Let  $Y_1$  and  $Y_2$  indicate heads on two fair coin flips, which are independent (Example 0.1), and let  $X = Y_1 + Y_2$  be the number of heads. Given  $X = 1$ , which has probability  $1/2$ , exactly one coin is heads, so:

$$\begin{aligned}\Pr(Y_1 = 1, Y_2 = 1 \mid X = 1) &= \frac{\Pr(Y_1 = 1, Y_2 = 1, X = 1)}{\Pr(X = 1)} && \text{(definition of conditional probability)} \\ &= \frac{0}{1/2} && \text{(two heads make } X = 2, \text{ not } 1) \\ &= 0 && \text{(divide)}\end{aligned}$$

but

$$\begin{aligned}
\Pr(Y_1 = 1 \mid X = 1) &= \frac{\Pr(Y_1 = 1, X = 1)}{\Pr(X = 1)} && \text{(definition of conditional probability)} \\
&= \frac{\Pr(Y_1 = 1, Y_2 = 0)}{\Pr(X = 1)} && (Y_1 = 1 \text{ and } X = 1 \text{ means } Y_2 = 0) \\
&= \frac{1/4}{1/2} && \text{(substitute)} \\
&= \frac{1}{2} && \text{(divide)}
\end{aligned}$$

and likewise  $\Pr(Y_2 = 1 \mid X = 1) = 1/2$ , so the product of the conditional probabilities is  $1/4 \neq 0$ :  $Y_1$  and  $Y_2$  are independent, but not conditionally independent given  $X$ .

**Proposition 0.1** (Neither independence nor conditional independence implies the other). *There are random variables  $Y_1$ ,  $Y_2$ , and  $X$  such that  $Y_1$  and  $Y_2$  are **conditionally independent** given  $X$  but not **independent**, and there are random variables  $Y_1$ ,  $Y_2$ , and  $X$  such that  $Y_1$  and  $Y_2$  are independent but not conditionally independent given  $X$ .*

#### **i** Proof

*Proof.* Example 0.3 gives the first case, and Example 0.4 gives the second. □

**Example 0.5** (Wet roads, umbrellas, and rain). Consider Brian Hutchinson’s intuitive illustration of conditional independence (Hutchinson 2024):

Let  $w$  denote wet roads,  $u$  denote people carrying umbrellas, and  $r$  denote rain.

- **Marginally, wet roads and umbrellas are not independent** ( $w \not\perp u$ ). If you look out a window and see people holding open umbrellas, that increases the probability that the pavement outside is wet.
- **Conditionally on rain, they are independent** ( $w \perp u \mid r$ ). Once you know with certainty whether it is currently raining, learning whether pedestrians are carrying umbrellas tells you nothing more about the state of the roads:

$$\Pr(w, u \mid r) = \Pr(w \mid r) \Pr(u \mid r), \quad \Pr(w \mid u, r) = \Pr(w \mid r)$$

**Proposition 0.2** (Conditional independence when each  $Y_i$  depends only on its own  $X_i$ ). *Let  $\tilde{X} = (X_1, \dots, X_n)$  be a discrete random vector, and let  $Y_1, \dots, Y_n$  be **conditionally independent** given  $\tilde{X}$ . Suppose also that each  $Y_i$  depends on  $\tilde{X}$  only through  $X_i$ : for every  $\tilde{x} = (x_1, \dots, x_n)$  with  $\Pr(\tilde{X} = \tilde{x}) > 0$  and every set of real numbers  $A_i$ ,*

$$\Pr(Y_i \in A_i \mid \tilde{X} = \tilde{x}) = \Pr(Y_i \in A_i \mid X_i = x_i)$$

Then, for every such  $\tilde{x}$  and all sets of real numbers  $A_1, \dots, A_n$ :

$$\Pr(Y_1 \in A_1, \dots, Y_n \in A_n \mid \tilde{X} = \tilde{x}) = \prod_{i=1}^n \Pr(Y_i \in A_i \mid X_i = x_i)$$

### **i** Proof

*Proof.* Each conditional probability given  $X_i = x_i$  is defined, because  $\Pr(X_i = x_i) \geq \Pr(\tilde{X} = \tilde{x}) > 0$ . So:

$$\begin{aligned} \Pr(Y_1 \in A_1, \dots, Y_n \in A_n \mid \tilde{X} = \tilde{x}) &= \prod_{i=1}^n \Pr(Y_i \in A_i \mid \tilde{X} = \tilde{x}) \quad (\text{conditional independence given } \tilde{X}) \\ &= \prod_{i=1}^n \Pr(Y_i \in A_i \mid X_i = x_i) \quad (\text{each } Y_i \text{ depends on } \tilde{X} \text{ only through } X_i) \end{aligned}$$

□

*Remark.* In regression,  $\tilde{X}$  is usually the full set of covariates  $(X_1, \dots, X_n)$ , and a model usually also assumes, as Proposition 0.2 does, that each  $Y_i$  depends on the covariates only through its own  $X_i$ . That second assumption is a separate one, not part of conditional independence.

**Definition 0.3** (Identically distributed). Random variables  $X_1, \dots, X_n$  are **identically distributed** if they all have the same [CDF](#):

$$\forall i \in \{1, \dots, n\}, \forall x \in \mathbb{R} : \Pr(X_i \leq x) = \Pr(X_1 \leq x)$$

**Theorem 0.3** (Identically distributed discrete random variables have the same PMF). Discrete random variables  $X_1, \dots, X_n$  are *identically distributed* if and only if they all have the same [PMF](#): for every  $i \in \{1, \dots, n\}$  and every real number  $x$ ,

$$\Pr(X_i = x) = \Pr(X_1 = x)$$

## i Proof

*Proof.* Write  $F_i(t) \stackrel{\text{def}}{=} \Pr(X_i \leq t)$  for the CDF of  $X_i$ .

**Only if: the CDF determines the PMF through its jumps.** Fix  $i$  and  $x$ . Every number below  $x$  lies in exactly one of the intervals  $(-\infty, x-1]$ ,  $(x-1, x-\frac{1}{2}]$ ,  $(x-\frac{1}{2}, x-\frac{1}{3}]$ , ..., so the event  $\{X_i < x\}$  is the disjoint union of the events  $B_1 \stackrel{\text{def}}{=} \{X_i \leq x-1\}$  and  $B_k \stackrel{\text{def}}{=} \{x - \frac{1}{k-1} < X_i \leq x - \frac{1}{k}\}$  for  $k \geq 2$ , and for each  $K$ , the events  $B_1, \dots, B_K$  have union  $\{X_i \leq x - \frac{1}{K}\}$ :

$$\begin{aligned} \Pr(X_i < x) &= \sum_{k=1}^{\infty} \Pr(B_k) && \text{(countable additivity)} \\ &= \lim_{K \rightarrow \infty} \sum_{k=1}^K \Pr(B_k) && \text{(definition of an infinite series)} \\ &= \lim_{K \rightarrow \infty} \Pr(X_i \leq x - \frac{1}{K}) && \text{(finite additivity)} \\ &= \lim_{K \rightarrow \infty} F_i(x - \frac{1}{K}) && \text{(definition of the CDF)} \end{aligned}$$

The event  $\{X_i \leq x\}$  is the disjoint union of  $\{X_i < x\}$  and  $\{X_i = x\}$ , so:

$$\begin{aligned} P(X_i = x) &= \Pr(X_i \leq x) - \Pr(X_i < x) && \text{(finite additivity)} \\ &= F_i(x) - \lim_{K \rightarrow \infty} F_i(x - \frac{1}{K}) && \text{(definition of the CDF)} \\ &= F_i(x) - \lim_{K \rightarrow \infty} F_i(x - \frac{1}{K}) && \text{(the display above)} \\ &= F_1(x) - \lim_{K \rightarrow \infty} F_1(x - \frac{1}{K}) && (F_i = F_1) \\ &= P(X_1 = x) && \text{(the same two steps, for } X_1) \end{aligned}$$

**If: the PMF determines the CDF through sums.** Let  $S$  be the set of values  $u$  with  $P(X_1 = u) > 0$ . Since the PMFs are equal,  $S$  is also the set of  $u$  with  $P(X_i = u) > 0$ , so  $S \subseteq \mathcal{R}(X_i)$  and  $S \subseteq \mathcal{R}(X_1)$ . For every real  $t$ , the event  $\{X_i \leq t\}$  is the disjoint union of the events  $\{X_i = u\}$  over the countably many  $u \in \mathcal{R}(X_i)$  with  $u \leq t$ , so:

$$\begin{aligned} F_i(t) &= \sum_{u \in \mathcal{R}(X_i), u \leq t} P(X_i = u) && \text{(countable additivity)} \\ &= \sum_{u \in S, u \leq t} P(X_i = u) && \text{(drop the terms that are 0)} \\ &= \sum_{u \in S, u \leq t} P(X_1 = u) && \text{(the PMFs are equal)} \\ &= \sum_{u \in \mathcal{R}(X_1), u \leq t} P(X_1 = u) && \text{(restore the terms that are 0)} \\ &= F_1(t) && \text{(countable additivity)} \end{aligned}$$

□

*Remark.* As with independence (Example 0.2), the PMF form would not work as a definition in general: every continuous random variable has  $\Pr(X_i = x) = 0$  at every  $x$ .

**Example 0.6** (Identically distributed but not independent). In Example 0.1,  $X_1$  and  $1 - X_1$  (the indicator of tails on the first flip) both take the values 0 and 1 with probability  $1/2$  each, so they are identically distributed (Theorem 0.3). They are not independent: knowing one determines the other.

**Definition 0.4** (Conditionally identically distributed). Random variables  $Y_1, \dots, Y_n$  are **conditionally identically distributed** given random variables  $X_1, \dots, X_n$  if the conditional CDF of each  $Y_i$  given  $X_i = x$  is one shared function  $G(y | x)$  of  $y$  and  $x$ : for every  $i \in \{1, \dots, n\}$ , every  $y$ , and every  $x$  with  $\Pr(X_i = x) > 0$ ,

$$\Pr(Y_i \leq y | X_i = x) = G(y | x)$$

When  $X_i$  is continuous, the condition applies at every  $x$  with  $p(X_i = x) > 0$ , with  $\Pr(Y_i \leq y | X_i = x)$  computed from densities: the integral of  $p(X_i = x, Y_i = u) / p(X_i = x)$  over  $u \leq y$  (a sum over values  $u \leq y$  when  $Y_i$  is discrete).

**Example 0.7** (A shared regression model). Suppose each  $Y_i$  is binary, with  $\Pr(Y_i = 1 | X_i = x) = \pi(x)$  for one function  $\pi$  shared by every  $i$  (for instance,  $\pi(x) = x/(1+x)$  for a dose  $x \geq 0$ ). Then  $Y_1, \dots, Y_n$  are conditionally identically distributed given  $X_1, \dots, X_n$ , with  $G(y | x) = 1 - \pi(x)$  for  $0 \leq y < 1$  (and 0 for  $y < 0$ , 1 for  $y \geq 1$ ). Their marginal distributions can still differ: if participant 1 always receives dose 0 and participant 2 always receives dose 1, then  $\Pr(Y_1 = 1) = 0$  but  $\Pr(Y_2 = 1) = 1/2$ .

**Definition 0.5** (Independent and identically distributed). Random variables  $X_1, \dots, X_n$  are **independent and identically distributed** (shorthand: “ $X_i$  iid”) if they are:

- [statistically independent](#), and
- [identically distributed](#).

*Remark.* The IID assumption is one of the most common assumptions in introductory statistics: it says a sample  $X_1, \dots, X_n$  can be treated as  $n$  independent draws from a single shared distribution.

**Example 0.8** (Repeated die rolls). The results of  $n$  rolls of the same fair die are IID: the rolls are independent, and each is uniform on  $\{1, \dots, 6\}$ .

**Definition 0.6** (Conditionally independent and identically distributed). Random variables  $Y_1, \dots, Y_n$  are **conditionally independent and identically distributed** given  $X_1, \dots, X_n$  (shorthand: “ $Y_i \mid X_i$  ciid” or just “ $Y_i \mid X_i$  iid”) if:

- $Y_1, \dots, Y_n$  are **conditionally independent** given  $(X_1, \dots, X_n)$ ,
- each  $Y_i$  depends on  $(X_1, \dots, X_n)$  only through  $X_i$ , and
- $Y_1, \dots, Y_n$  are **conditionally identically distributed** given  $X_1, \dots, X_n$ .

**Example 0.9** (The usual regression assumption). In Example 0.7, if the  $Y_i$  are also conditionally independent given all the doses, and each  $Y_i$  depends on the doses only through its own  $X_i$ , then  $Y_i \mid X_i$  ciid, and the joint conditional PMF is  $\prod_{i=1}^n \pi(x_i)^{y_i} (1 - \pi(x_i))^{1-y_i}$ : one shared function evaluated at each  $(x_i, y_i)$ .

 Video lecture

Hutchinson’s [Probability Refresher](#) (27 min) covers independence (Hutchinson, n.d.). The login for the video site is posted [on Canvas](#).

## References

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